Autodesk®
Scaleform®

ActionScript 2 XML User Guide

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Autodesk® Scaleform® 4.3

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Autodesk Scaleform SDK Flash® PC Adobe® Flash® Studio Scaleform

Flash ActionScript 2 XML AS2 XML API XML XMLNode DOM XML Scaleform 4.1 Scaleform DOM SAX2 XML Scaleform Player Expat ... Scaleform XML :Windows® Linux® MacOS® Xbox 360® PSP® (PlayStation®Portable) PlayStation®2 (PS2) PlayStation®3 (PS3™) Wii®

Scaleform XML

2 XML

ゟ ﾉ´ ヽヤヽヶ ヽṇ ヶ ヽヤヽヽ` ¡ ヶṇ ヽṇ ﾅ Scaleform XML ヤヤヽヶヤヽṇヽヶヤ~ ļ
. ∩´ ﾝ ﾉ ~ﻲﻟﺋ´ ヽ ヶ~ ﾅ ﾁ ~ ~ Ã

2.1 ActionScript の XML サポートを有効にする

ActionScript XML ピ ヸ ヽヤṇヽ~ Ã ゲ´ ~. ´ ¡ Scaleform n ゔ ă ゙ ゎ` XML ヽṇヽṇ ヽヤṇヽ~ Scaleform ヶṇヽṇ ï´ ˆ ﾉ~. ゚ Í ﾅ ゼ ~ ~ Ã ゟ ﾉ ¡ヽヤヽヶヸ FxPlayer.cpp ヤı ヶ ﾌ. ゙ ˘ ï` ~ Ã ﾉ ﾍ ゙ ヽヤṇヤ~ ヽヤヤヶ ヽṇヽṇ ヤヤヤヶヤヽṇヽヶヤ´. ⅞… ï` ク ~ ~ :

…
```
#include "XML/XML_Expat.h"
using namespace Scaleform;
…
Ptr<XML::Parser> pexpatXmlParser = *new XML::ParserExpat;
Ptr<XML::SupportBase> pxmlSupport = *new XML::Support(pexpatXmlParser);
mLoader.SetXMLSupport(pxmlSupport);…
```

2.2 C++言語のドキュメント解析

ヽṇヽṇ ヤヶṇヶ~ ⅞… ¨ ˘ ¡ C++ ﾌ. n ى XML ヶヽヶヶヤヶ~ ゚ ﾇ~. ゔ` ﾅ` ク~ ~ Ã
ActionScript XML ヽヤṇヶﾅ Ã ゲ` ゼ ~ ¡ ゟ ﾉ ヽṇ ヶ~ ⅞… ï` ˘ XML ヤı ヶ ﾌ. DOM ヶ ヽṇ~ ð¤ ` ク~ ~ (Scaleform ヽṇヤ SDK ヶ ヽヤヤﾅ ゚ Í ` ˘) Ã

…
```
#include "GFx/XML/XML_DOM.h"
using namespace Scaleform;
…
// ピ ﾋ ヶ ピ ă~.  DOM ヽ ヶ ヤ~ ð¤ ~.
XML::DOMBuilder domBuilder(Loader.GetXMLSupport());
// ~ ゚  ¡ ピ ﾋ ヶ ピ ゙ ~.  DOM ヽ ヶ ヤ~ ð¤ ~.
XML::DOMBuilder domBuilder2(Loader.GetXMLSupport(), true);
…
// ゔ xmlヤı ヶ~ ピ ă` ˘ ¡ DOMヶヽṇ ヶṇヶ~ ゃ~
// (ﾊ ﾉ` ヽヤヤ ヽヶ ∪ヽṇヤヤ~ ð¤ ~. )
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                                mLoader.GetFileOpener());
// ~ ゚  ¡ ゔ xmlヤı ヶ~ ピ ă` ˘ ¡ DOMヶヽṇ ヶṇヶ~ ゃ~ (ㄣ ゥ ﾍ  ` ヽヤヤ ヽヶ ∪ヽṇヤヤ~ ⅞… ï~. )
Ptr<XML::ObjectManager> pobjMgr = *new XML::ObjectManager();
```

```
Ptr<XML::Document> pdoc2 = domBuilder.ParseFile("inputfile.xml",
                                                mLoader.GetFileOpener(), pobjMgr);
…
```

ActionScript XML ヽ ヽ ṇ ヽ ュ Ả ゲ ˋ ˊ ヿ ㇾ ㇵ ị ヽ ṇ ヽ ṇ̓ ヽ ヽ ㇾ 1̈ ㇵ ㇾ ˇ ㇾ ヽ ヽ ㇾ ヽ ~ ðȣ ˋ ị ㇿ ㇷ . ㇿ ´ ⅞… ĩ ˜ . ㇽ ˋ ュ ˋ ᵱ ˜ ˘ :

```
…
#include "GFx/XML/XML_DOM.h"
#include "GFx/XML/XML_Expat.h"
using namespace Scaleform;
…
Ptr<XML::Parser> pexpatXmlParser = *new XML::ParserExpat;
Ptr<XML::Support> pxmlSupport = *new XML::Support(pexpatXmlParser);
…
// پ إ ゲ ピ ắ ˜ . DOM ヽ ㇵ ヽ ~ ðȣ ˘ .
XML::DOMBuilder domBuilder(pxmlSupport);
// پ إ ゲ ピ ぃ ˜ . DOM ヽ ㇵ ヽ ~ ðȣ ˘ .
XML::DOMBuilder domBuilder2(pxmlSupport, true);
…
// ㇿ ヽ ヽ ヽ ㇾ ピ ắ ㎡
…
```

2.3 C++言語で DOM ツリーをトラバースする

ㇿ ㇿ ヽ ṇ ㇵ ` DOM ㄱ ヽ ṇ̓ ㇳ ㅂ ュ ㇳ ㇷ ㇰ ˜ ˘ (Scaleform ヽ ṇ ヽ SDK ヶ ヽ ㇾ ヽ ュ ˚ í ` ˘):

```
…
using namespace Scaleform;
…
void PrintDOMTree(XML::Node* proot)
{
    switch (proot->Type)
    {
        case XML::ElementNodeType:
        {
            XML::ElementNode* pnode =
                static_cast< XML::ElementNode*>(proot);
            if (pnode->Prefix.GetSize() > 0)
            {
                printf("ELEM - '%s:%s' ns:'%s' prefix:'%s'"
                       " localname: '%s'",
                       pnode->Prefix.ToCStr(),
                       pnode->Value.ToCStr(),
                       pnode->Namespace.ToCStr(),
```

```
                    pnode->Prefix.ToCStr(),
                    pnode->Value.ToCStr());
            }
            else
            {
                printf("ELEM - '%s' ns:'%s' prefix:''"
                    " localname: '%s'",
                    pnode->Value.ToCStr(),
                    pnode->Namespace.ToCStr(),
                    pnode->Value.ToCStr());
            }
            for (XML::Attribute* attr = pnode->FirstAttribute;
                  attr != NULL; attr = attr->Next)
            {
                printf(" {%s,%s}", attr->Name.ToCStr(),
                        attr->Value.ToCStr());
            }
            printf("\n");
            for (XML::Node* child = pnode->FirstChild; child != NULL;
                  child = child->NextSibling)
                PrintDOMTree(child);
            break;
        }
        case TextNodeType:
        {
            printf("TEXT - '%s'\n", proot->Value.ToCStr());
            break;
        }
        default:
        {
            printf("UNKN\n");
        }
    }
}
…
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                                Loader.GetFileOpener());
PrintDOMTree(root);
```

2.4 デバッグレポートの設定

Scaleform ヽ ṇ ヤ ヤ ı ̃ ｲ ´ ̄ ヽ ヒ ヤ ̃ . ｲ ṇ ヽ ṇ̉ ｉ ợ Ọ õ ̂ ж ќ ỏ Ớ Ọ ڳ ｲ ヒ ṇ ヶ ヤ ヤ ヽ ̃ L ﾒ ̃ . ` . XML_DOM.cpp ´ ヤ ヤ ヽ ̃ ̃ ﾒ / ̂ ض ̃ . ↄ ` ﾚ ` ｐ ̃ ̃ ﾑ

```
// ヽ ヒ ヽ ヽ ヤ ﾉ ヤ ヤ ヽ & ヽ ｲ ṇ ヤ ヤ ヤ ヽ
//
#ifdef SF_BUILD_DEBUG
//
// ヽ ヒ ヽ ヽ ヤ ﾉ ̉ ｲ ヽ ｲ ヽ ヒ ヽ ヶ ｲ ヶ̉ DOM ｲ ヽ ṇ ڳ ﾂ ̃ ヽ ｲ ṇ ヤ ̃ .
//
// #define SF_XML_DOCBUILDER_TRACE
//
// ðȣ ヒ ` DOM ｲ ヽ ṇ ̃ ̃ ̀ ̆ غ ى ヤ ﾉ ´ ヤ ヒ ヒ ̃ .
// (ق փ : ヶ ｐ ´ ヤ ı ̃ ｲ ̉ ヒ ｲ ̉ ↄ ̣ ̃ ﻡ i ̣ ↄ ` )
//
// #define SF_XML_DOCBUILDER_DOM_TREE_DUMP
//
#endif  // #ifdef SF_BUILD_DEBUG
```

3

Scaleform expat XML GFx::XML::Parser XML_Parser.h GFx::XML::ParserHandler SAX2

```
namespace Scaleform { namespace GFx { namespace XML {
//
// XML
//
//
//
//
// XML
//
//
class Parser : public RefCountBaseNTS<Parser, Stat_Default_Mem>
{
  public:
    virtual ~Parser() {}

  //
    virtual bool    ParseFile(const char* pfilename, FileOpenerBase* pfo,
                              ParserHandler* pphandler) = 0;
    virtual bool    ParseString(const char* pdata, UPInt len,
                                ParserHandler* pphandler) = 0;
};
}}} //SF::GFx::XML
```

XML ParseFile ParseString DOM GFx::XML::ParserLocator GFx::XML::ParserHandler

```
namespace Scaleform { namespace GFx { namespace XML {
//
// SAX2
//
// DOM SAX2
//
//
class ParserHandler : public RefCountBase<ParserHandler, StatMV_XML_Mem>
```

```
{
  public:
    ParserHandler() {}
    virtual ~ParserHandler() {}

  // ١ ٦ ١ ٢ ٤ ٧ ےپ` ڴك
    virtual void        StartDocument() = 0;
    virtual void        EndDocument() = 0;

  // ٧ ٧ ٦ ١ ٢ ٤ ٧ ےپ` ڴك
    virtual void        StartElement(const StringRef& prefix,
                                     const StringRef& localname,
                                     const ParserAttributes& atts) = 0;
    virtual void        EndElement(const StringRef& prefix,
                                   const StringRef& localname) = 0;

  // U ٨ پ ٨ ش ٥ ٱ ٿ ٦ ١ ٢ ٤ ٧ ز ڎ ٤ ٧ ٧ ٤ ٧ ۏ ` ´ . ٱ
    virtual void        PrefixMapping(const StringRef& prefix,
                                      const StringRef& uri) = 0;

  // ٧ ٧ ٦ ١ ٢ ٤ ٧ ٨ ٦ ٦ ٦ ٦ ٦ ٿ ٧
    virtual void        Characters(const StringRef& text) = 0;

  // پ إ ك گ
    virtual void        IgnorableWhitespace(const StringRef& ws) = 0;

  // ۽ ت ڏ ٦ ١ ٢ ٤ ٧
    virtual void        SkippedEntity(const StringRef& name) = 0;

  // GFx::XML::Parser ٤ ٤ ٧ ٢ ٤ ٧ ز ڃ ڈ ٦ ٿ ١ ٤ ٧ ٦ ٧ ڤ ف ڀ ٨ ´ ز
  // ١ ٦ ١ ٢ ٤ ٧ ١ ٤ ٦ ٿ ٧ ~ ˜ ٨ ˜ . ˚ ٱ ڀ ط ٱ GFx::XML::ParserHandler ٤ ٤ ٧ ٢ ٤ ٧ ٿ ٦ ١ ٤ ڃ ز
  // ٦ ٧ ٿ U ظ ` ز ٦ ٤ ٧ ٿ ٧ ٤ ٧ ز xmlversion ز ٨ ´ ٧ ٧ ٤ ١ ¯ ١ ٤ ٤ ١ ٢ ٤ ٧ ˜
  //ˆ ` ש ت ڈ ˜ . ` ´ ز ١ ٦ ٿ ٧ ٦ ٤ ٧ ˆ ٧ ٧ ڀ ٤ ˚ ٱ ` ط ٱ
    virtual void    SetDocumentLocator(const ParserLocator* plocator) = 0;

  // ٧ ٢ ٤ ٧
    virtual void        Comment(const StringRef& text) = 0;

  // ErrorHandler ٧ ٿ ١ ٤ ٧ ٧
    virtual void        Error(const ParserException& exception) = 0;
    virtual void        FatalError(const ParserException& exception) = 0;
    virtual void        Warning(const ParserException& exception) = 0;

};
}}} //SF::GFx::XML
```

4

4.1 ActionScript にカスタム *XML* ノード プロパティが存在しない

```
// ﻥ ﻥ ı ﻧ ﻧ ﻣ˜ ﻫ˘
temp = temp.prevSibling;
// ﻥ ﻣ ﻣ ﻣ̊ ﻳﻢ ˜ ﺿ˘
trace(temp.attributes.someProperty)
```

Scaleform ` someValue ﺯ ﻡ ﻝ ﻱ ˜ ˘ ً

4.2 ActionScript XML.status プロパティが存在しない

ﻝ ﻣ ﻣ ﻣ̊ ﻧ ﻥ ﻧ ﻥ ﻣ˝ ﻧ ﻧ ﻣ̊ ﻝ ﻧ ﻥ ﻧ ﻥ 1˜ ¡ ActionScript 2.0 XML˚ ﻝ ﻧ ﻥ ﻧ ﻥ 1´ ﻙ ﻥ ﻣ ﻧ ﻧ˜. ` ﻕ´ ﻱ˜´ ˜ ´ ŭ ﻑ ﻫﺷﻢ ă̈ ﺯ ﻍ ﻍ˜. ` ¡ ﺟ̊ ﻧ ﻓ ﻧ ﻣ˝˚ ă̈ ﻑﺦ ﻱ ˜ ,˜ ` ` ` ً

ActionScript ﻝ ﻧ ﻥ ﻧ ﻥ 1:

0 ﻝ ﻧ ﻥ´ ` ; ˚ ﺟ̊ ﺯ δ´ ﺵ ﺝ ﻳﻪ ˜ ` ` ً

-2 CDATA ﻧ ﻥ ﻥ ﻓ ﻧ ﺯ .. ﺉ´ ﻙ ﻳﻪ ˘ ﻉ˜ ,˜ ً

-3 XML ﺵ ﻇ ﺯ .. ﺉ´ ﻙ ﻳﻪ ˘ ﻉ˜ ,˜ ً

-4 DOCTYPE ﺵ ﻇ ﺯ .. ﺉ´ ﻙ ﻳﻪ ˘ ﻉ˜ ,˜ ً

-5 ﻧ ﻣ ﻧ ﻣ ﺯ .. ﺉ´ ﻙ ﻳﻪ ˘ ﻉ˜ ,˜ ً

-6 XML ﻝ ﻓ ﻣ ﻧ ﻣ̊ ﺡ ﻫ´ ﺯ δ ﺯ ﻁ ˜ ` ` ً

-7 ﻣ ﻣ ﻣ ﺯ ŭ ﺽ ˘ ˘ ﻉ˜ ˜ ً

-8 ﻳﻢ ﮋ ﺯ .. ﺉ´ ﻙ ﻳﻪ ˘ ﻉ˜ ,˜ ً

-9 ˘ ا ﻣ ﻣ ` ﻙ ﻳﻪ ﻣ ﻣ ﺯ ﻫ ﻫﻢ ˜ ,˜ ` ` ` ً

-10 ﻫ ﻫﻢ˜. ˘ ا ﻣ ﻣ̊ ´ ﻉ ﻙ ﻳﻪ ﻣ ﻣ ﺯ ﻣ ﻫ ﻱ ˜ ` ` ً

Expat ﭖ ActionScript ` ˚ ﻟ ﻣ ﻫ ﻧ ﻣ Ķ

```
//
// XML_ERROR_NONE           = 0
// XML_ERROR_INVALID_TOKEN  = 0 (例: XML.parse("http://www.google.com"))
// XML_ERROR_UNCLOSED_CDATA_SECTION = -2
// XML_ERROR_UNCLOSED_TOKEN = -3
// XML_ERROR_INVALID_TOKEN  = -4
// XML_ERROR_INVALID_TOKEN  = -5
// XML_ERROR_UNCLOSED_TOKEN = -8
// XML_ERROR_NO_ELEMENTS    = -9
// XML_ERROR_TAG_MISMATCH   = -10
//
```

ヿヤṇ ヤṇ 1˜ 1ㅅ1` レヽヤヤレヤ`´ i˙ ˜ i˙ ㅓ ₆ ゚ ゐ ヤヤṇヤヤ ヤヰヤヤ˝ ˜ ǟ ﬠ˜˙ ↄ `゚ ` ק˜ ,˜ Ĩ *Scaleform::XML::DOMBuilder* ㅓ´ ヿヤṇ ヤヽヤṇヤ˜ ŭ ȣ` i XML ヤṇヤ ṇ˝ ヤヤヤヤヤ´. ¨ ˇ ط ﬡ ㄟ ˇ ך *GFx::XML::ParserLocator* ヽヤヤ˝ ヤヅ ヤṇヤ˜ ⅞… Ĩ` ˇ i˙ ヤ ㄡ˜ ヤヤヽヤ`˜ ˜ Ĩ

4.3 ActionScript XML.docTypeDecl プロパティが存在しない

Flash® ㅓ ǎ ǟ ج . : ṡ ActionScript XML ヤṇヤṇ̓ i˙ ﾟ₆ ヮ g ĩ ヤṇヤṇ`゚ ゐ ˜ ¸˜ Ĩ DOCTYPE ش ᵾ ヤṇヤṇ´. . ﺝ˜ ŭ. . i XML.docTypeDecl ヤヰヤヤ˝ ´ Ĩ ﺝ ㄟ ˜ ˜ ﺯ i DTD ﾟ₆ ヮ゚ ヵ . ˜ ¸˜ Ĩ Ş ↄ̓ ヤヰヤヤ˝゚ Flash´. Scaleform´. ˚ Í´ ך̓ ` i ك ض ㄟ ˇ ך˜ ˜ Ĩ